

3.3

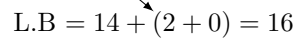
5. Every month a plastics plant must make batches of five different types of plastic toys. There is a conversion cost c_{ij} in switching from the production of toy i to toy j , shown in the matrix. Find a sequence of toy production (to be followed for many months) that minimizes the sum of the monthly conversion costs.

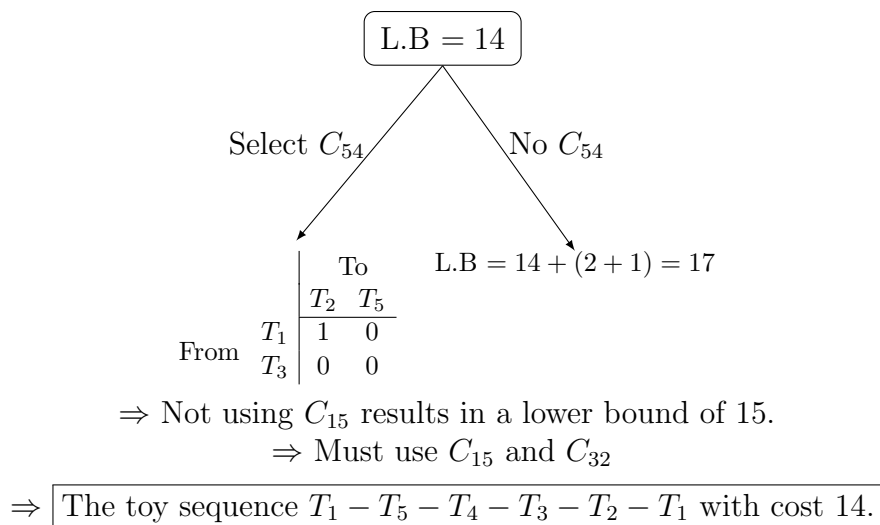
		To				
		T_1	T_2	T_3	T_4	T_5
From	T_1	—	3	2	4	3
	T_2	4	—	4	5	5
	T_3	5	3	—	4	4
	T_4	3	5	1	—	6
	T_5	5	4	2	3	—

* Subtract each entry by the minimum value of each row i and column j

		To				
		T_1	T_2	T_3	T_4	T_5
From	T_1	∞	3	<u>2</u>	4	3
	T_2	<u>4</u>	∞	4	5	5
	T_3	5	<u>3</u>	∞	4	4
	T_4	3	5	<u>1</u>	∞	6
	T_5	5	4	<u>2</u>	3	∞
						<u>12</u>

		To				
		T_1	T_2	T_3	T_4	T_5
$\Rightarrow$ From	T_1	∞	1	0	2	<u>1</u>
	T_2	0	∞	0	<u>1</u>	1
	T_3	2	0	∞	1	1
	T_4	2	4	0	∞	5
	T_5	3	2	0	1	∞
		0	0	0	1	1
						2

$$\text{L.B} = 12 + 2 = 14$$




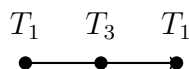
7. (a) Use the approximate algorithm to find approximate traveling salesperson tours for the cost matrix in Exercise 5 (use just entries above the main diagonal).

		To				
		T_1	T_2	T_3	T_4	T_5
From	T_1	∞	3	2	4	3
	T_2	3	∞	4	5	5
	T_3	2	4	∞	4	4
	T_4	4	5	4	∞	6
	T_5	3	5	4	6	∞

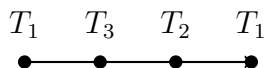
- Select T_1



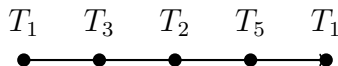
- Add T_3 to the circuit C (T_1 is 2 units away from T_3)



- Add T_2 to C (T_2 is 3 units away from T_1)



- Add T_5 to C (T_5 is 3 units away from T_1)



- Add T_4 to C (T_4 is 3 units away from T_3)

